

Shubham Kumar

Lead SDET (AI & Automation)

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Profile Summary

Dynamic and detail-oriented SDET Leader with 7+ years of expertise in architecting scalable test automation frameworks and driving quality-centric software delivery. Currently specialized as a Lead SDET (AI & Automation), focusing on Agentic AI testing and LLM-driven workflows. Expert in leveraging LangChain to automate complex validation, optimizing CI/CD pipelines, and leading high-performing teams to enhance product reliability in high-throughput environments.

Technical Skills

AI & ML	Agentic AI, LangChain, GPT API Validation, Prompt Engineering, LLM Observability, RAG Validation
Languages	Java, JavaScript, Python, C++, SQL
Automation	Selenium, Cypress, Playwright, Appium, Robot Framework, Pytest, Rest Assured
Backend	Spring Boot, Spring Data JPA, Node.js, Kafka, Redis, Microservices Architecture
DevOps	AWS (S3, SNS, SQS, Lambda, EC2), Docker, Jenkins, GitHub Actions, Bitbucket
Databases	MySQL, MongoDB, PostgreSQL, Oracle, Redis
Tools	JMeter, Locust, Cucumber, Gherkin, Jira, Azure DevOps, Allure, Postman

Experience

- Oct 2025–Pres. **Lead SDET (AI & Automation)**, *Times of India*, Bangalore
- Architected an Agentic AI testing framework using LangChain to automate the validation of personalized music streaming applications.
 - Developed intelligent test agents that simulate user behavior to discover edge cases in AI-driven music recommendation engines.
 - Implemented automated evaluation metrics for LLM-generated playlists, ensuring high relevance and metadata accuracy.
 - Optimized backend validation for music content delivery networks using Python and LangChain-based verification scripts.
 - Established LLM observability pipelines to monitor model drift and hallucination rates in production-like test environments.
- Apr 2025–Oct 2025 **SDET Manager**, *BestBuy*, Bangalore
- Architected end-to-end automation strategies for Java Spring Boot microservices, leading a regression initiative that reduced testing efforts by 45%.
 - Engineered multi-threading execution which reduced the total automated suite execution time by 65%.
 - Collaborated with Product and DevOps teams to define Acceptance Criteria (BDD/Gherkin) ensuring quality is built-in from the sprint start.
 - Directed performance testing and monitoring efforts utilizing JMeter and AWS CloudWatch to ensure system resilience.
 - Designed and implemented automated testing frameworks for APIs (Rest Assured) and Web UI (Selenium) integrated within Jenkins pipelines.

- Feb 2024–Feb 2025 **Lead SDET**, *Probe 42*, Bangalore
- Led QA strategy for LLM-based applications, building custom Robot Framework libraries for UI testing and Pytest frameworks for backend GPT API validation with mocking support.
 - Built and managed CI/CD pipelines using Git, Jenkins, Docker, and AWS EC2, automating daily builds and deployments to accelerate delivery cycles.
 - Conducted load, stress, and performance testing using JMeter and Django ORM, ensuring database integrity and system resilience under high traffic.
 - Implemented automated prompt engineering validation suites to test LLM robustness against adversarial inputs and prompt injection.
 - Developed custom LangChain-based agents to perform autonomous exploratory testing of complex data retrieval and RAG workflows.
- Sep 2022–Feb 2024 **SDET**, *GradRight*, Hyderabad
- Developed front-end automation frameworks from scratch using Cypress and Playwright with Java and JavaScript.
 - Performed deep database testing in MongoDB, writing complex queries for data integrity and validation.
 - Optimized 500+ API Automation tests with Rest Assured and Java, validating complex integration backend workflows.
 - Designed and executed a comprehensive POC using WebdriverIO and Mocha, automating 200+ test cases with dynamic locator strategies.
 - Introduced retry logic in Selenium and Appium-based frameworks to stabilize flaky tests and improve pipeline consistency.
- Oct 2021–Sep 2022 **SDET**, *Info Edge India Ltd.*, Noida
- Automated REST API testing for microservices, increasing overall API test efficiency by 40%.
 - Built Cypress and Playwright automation frameworks from scratch, improving UI test coverage by 30%.
 - Implemented BDD using Cucumber and Gherkin, reducing defect turnaround time by 25%.
 - Managed environment-specific configurations with Bitbucket and AWS, improving deployment reliability by 35%.
 - Integrated Jenkins with Allure and JUnit for automated test reporting and increased visibility.
- Apr 2019–Sep 2021 **QA & Automation Engineer**, *Sysnius Infotech*, Delhi
- Reduced manual testing efforts by over 50% by transitioning legacy test cases to automated suites.
 - Led a team of 4 engineers to develop foundational automation testing frameworks for client projects.
 - Owned the software lifecycle processes, leading to a 15% improvement in release quality and stability.
 - Performed backend and database testing across MySQL and PostgreSQL, writing SQL scripts for data validation.
 - Awarded 'Best Testing Engineer of the Month' in 2020 for contributions to product quality.

Achievements

- AI Innovation Pioneered the use of LangChain for automated UI exploration and music metadata validation.
- Leadership Awarded Employee of the Month at BestBuy for leadership in regression automation.
- Efficiency Successfully reduced release cycle times by 30% through CI/CD pipeline optimization.
- Mentorship Mentored 10+ junior and offshore team members, boosting team productivity by 20%.

Certifications

LangChain for LLM Application Development
 DevOps Certified DevOps Engineer - Edureka
 Automation Certified Automation Tester (Selenium) - Udemy
 Scrum Certified Scrum Master - Udemy

Education

- 2019 **B.Tech in Electronics & Telecommunications**, *Trident Academy of Technology*, BPUT,
6.86 CGPA
- 2015 **12th Grade**, *JAC Board*, 68.2%